

PALLAVI MARALLA SATISH

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Summary

Backend-focused Software Engineer owning production services across streaming, media, and research-platform domains. Experience spans Java, Node.js, and Python backend systems, AWS-to-GCP cloud migration, and end-to-end platform ownership, plus hands-on LLM integration and prompt-engineered pipelines. MS in Computer Science from Stevens Institute of Technology.

Experience

Full Stack Developer

June 2025 – May 2026 | Hoboken, New Jersey, USA

Systems Engineering Research Center (SERC) — DoW-Sponsored University Affiliated Research Center

- Architected and engineered an internal workflow automation platform from the ground up, as measured by adoption across 500 users spanning SERC/AIRC staff, by independently owning system design, custom application logic, and deployment with no predefined spec.
- Built custom backend logic beyond default no-code configuration, as measured by 6 forms, 10 views, and 15 reports powering the platform, by studying Zoho Creator's technical documentation to implement workflow rules, scripting, and data structures that replaced manual, email- and spreadsheet-based operations.
- Drove continuous evolution of the platform, as measured by repeated feature additions, retirements, and re-optimizations without a rebuild, by maintaining an extensible architecture that absorbed changing organizational requirements over time.
- Provided infrastructure support for the platform, as measured by reliable data backup and domain resolution, by provisioning and managing AWS S3 and Route 53.

SDE-I

April 2022 – July 2024 | Bangalore, Karnataka, India

Zee Entertainment Enterprises Limited — Media & Entertainment Conglomerate, Parent of the ZEE5 OTT Streaming Platform

- Led migration of Watch History and Launch API from AWS to Google Cloud Platform, as measured by personally executing approximately 95% of the technical migration work, by rewriting service configs and handling cutover across both systems.
- Modernized complex metadata queries, decreasing API response time by 35% through consolidating ten calls into two, allowing the API to scale to handle 5,000 req/s.
- Integrated the Music API with the Watch History service, as measured by a production cross-service endpoint layer, by designing and building the integration within a Java, ScyllaDB, and Confluent Kafka stack.
- Authored the first documentation for Watch History and Launch using Swagger/OpenAPI, as measured by adoption across other backend teams and cross-functional QA/frontend teams, by consolidating undocumented tribal knowledge into a shared reference.
- Owned production stability for Watch History and Launch API, as measured by resolving approximately 98% of bugs and Jira tickets raised against the services, by serving as first point of contact for incidents ahead of the tech lead and manager.

Android App Developer

March 2021 – May 2021 | Bellary, Karnataka, India

JSW Pvt Ltd — Steel and Infrastructure Conglomerate

- Built the MIS Generator Android application from scratch, as measured by delivering a fully functioning app as one of two interns with zero prior Android experience, by independently learning Kotlin development through documentation and self-directed research.

Projects

Query Plot

- Engineered a natural-language-to-chart pipeline, as measured by support for 7 distinct chart types generated without manual charting code, by building a prompt-engineered LLM integration (Ollama/CodeLlama) that converts plain-English queries into validated pandas/matplotlib/seaborn code.
- Architected secure execution for AI-generated code, as measured by eliminating direct filesystem and OS access at runtime, by building a sandboxed Python subprocess with a restricted builtins allowlist and automated code-sanitization before execution.

- Built a full-stack data visualization platform, as measured by 5 REST endpoints spanning upload, query, execution, download, and query-history, by implementing a React/TypeScript frontend against a Node.js/Express backend with PostgreSQL-backed history and Redis-based chart caching.

Technologies Used: React, TypeScript, Node.js, Express, PostgreSQL, Redis, Ollama/CodeLlama

GlowGraph - AI-Powered Skincare Storefront (Verrà)

- Engineered an agentic conversational AI system using Claude API with LangGraph-inspired state orchestration that classifies user intent (product search, refinement, cart actions) and autonomously routes queries to semantic search or direct recommendations, reducing response latency by 40% through elimination of redundant API calls.
- Built a hybrid semantic search pipeline, as measured by precise product-filter extraction from unstructured queries, by designing a constraint extraction layer that parses natural language into structured filters (price, concern tags, skin type) using local text-based embeddings with cosine similarity, eliminating dependency on external embedding APIs.
- Delivered a production-ready e-commerce platform, as measured by real-time cart synchronization and persistent order handling across concurrent sessions, by integrating Firebase multi-provider authentication (email/password, Google OAuth) with Redis-backed cart state and MongoDB order persistence across a 51-product catalog spanning 6 concern-based ranges.
- Architected a Docker-based microservices setup (Node.js/Express API, React frontend, MongoDB, Redis) with health-check orchestration and graceful shutdown handling, as measured by 95%+ accuracy on skincare-specific intent classification, by validating chat conversation quality through manual testing of query-routing correctness.

Technologies Used: Node.js, Express, React, Vite, Claude API, Firebase, MongoDB, Redis, Docker, Local Embeddings (Deterministic Hashing)

Education

Stevens Institute of Technology

September 2024 – May 2026

Master of Science in Computer Science

Coursework: Prompt Engineering, Applied AI / AI Technology Management, Agile Software Development, DevOps Principles

Dayananda Sagar College of Engineering

July 2018 – July 2022

Bachelor of Engineering in Computer Science and Engineering

Coursework: Software Development, Algorithm Design & Analysis, Object-Oriented Programming (Java), Database Management Systems (DBMS)

Skills

- **Languages:** JavaScript, TypeScript, Java, Kotlin, Python, C++, SQL, NoSQL, Bash Scripting
- **Backend:** Node.js, Express.js, Spring Boot, RESTful APIs, GraphQL, Microservices, API Gateway, API Versioning, API Monitoring, Load Balancing, Caching Strategies, Message Queues, Distributed Systems, Zoho Creator (custom logic & workflow automation)
- **Data:** PostgreSQL, Redis, ScyllaDB, MongoDB, MySQL, Amazon DynamoDB, Azure SQL, Data Pipelines
- **Cloud:** AWS (S3, Route 53, Lambda, API Gateway), Google Cloud Platform (service migration, Cloud Functions, Cloud Storage), Azure (Functions, Blob Storage, Key Vault), Serverless Architecture, Cloud-Native Development, Infrastructure as Code, Auto Scaling, Containerization
- **Security:** OAuth 2.0, JWT, AWS IAM, SSL/TLS, AWS KMS Encryption, HTTPS, Application Security, Data Encryption
- **DevOps:** Git, GitHub, GitLab, Bitbucket, Jenkins, AWS CodePipeline, AWS CodeBuild, Docker, Kubernetes, Terraform, CI/CD, Configuration Management, Monitoring and Logging
- **Messaging:** Apache Kafka (Confluent)
- **API Design:** Swagger/OpenAPI, Rate Limiting, Authentication, Authorization, API Security, API Testing, API Performance Optimization
- **Frontend:** React
- **AI/LLM:** Prompt Engineering, LLM Integration (Ollama/CodeLlama), Natural-Language-to-Code Pipelines, Sandboxed Code Execution